Modified SIPA Model

Juan Miguel Alberola Oltra, José Antonio Rodríguez Moreno,
Víctor Sánchez Anguix

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1 Introduction

This work presents a modified version of the original Student–Industry Placement Allocation (SIPA) model, which is primarily motivated by the explicit incorporation of academic supervisors into the allocation process. In its original formulation, the SIPA model considers only students and companies as decision-making agents, which may lead to solutions that are optimal from a computational perspective but insufficiently aligned with academic and organizational realities.

In real internship allocation processes, academic supervisors play an important role in the supervision and evaluation of internships. Ignoring their involvement may result in assignments that are difficult to manage in practice, either due to excessive supervision loads or inefficient geographic distributions of supervised placements. For this reason, the proposed modification introduces professors as a third agent in the model, with the aim of capturing additional academic and organizational criteria that influence feasible and desirable assignments.

At a conceptual level, professors are incorporated into the model as academic supervisors in order to reflect organizational and logistical aspects that are not captured when only students and companies are considered.

More specifically, distance-based supervision costs are incorporated in order to reflect the travel effort associated with overseeing geographically distributed internships, thereby discouraging solutions that concentrate highly dispersed placements under the same supervisor. Furthermore, an upper bound on the number of students that can be assigned to each professor is imposed, ensuring that supervision loads remain within manageable limits.

2 Model Overview and Modifications

This section provides an overview of the proposed modifications to the SIPA model, which are introduced in order to explicitly incorporate academic supervisors as structural agents in the allocation process. The inclusion of professors alters the original formulation by introducing supervision-related constraints and

cost components that must be jointly considered alongside student–company assignments.

In the original SIPA model, the allocation problem is defined exclusively in terms of students, internship positions, and companies. While this representation is sufficient to capture preference-based matching between students and industry, it does not account for the academic supervision layer that is essential in real-world internship programs. The proposed model addresses this limitation by introducing professors as supervisors whose role is to regulate the assignment process at a global level.

As a consequence, the modifications introduced in this work affect multiple components of the original SIPA framework, including:

- The set of agents explicitly represented in the model.
- The modeling of supervision-related costs associated with distance and dispersion.
- The structure of the objective function.
- The introduction of capacity constraints limiting the number of students supervised by each professor.

Therefore, the proposed formulation should be regarded as a modified version of the original SIPA model rather than a direct extension of its original structure.

In the proposed model, professors are formally introduced as academic supervisors whose impact on the allocation is represented through aggregated organizational and logistical criteria, including supervision-related cost components and capacity constraints.

A logistics-related component based on distance is incorporated to capture the supervision effort required by a given assignment. This effort is modeled in an aggregated manner, considering both the location of the educational center and the set of internships assigned to each professor. Assignments in which the supervised internships are excessively dispersed or located far from the supervisor are penalized, thereby favoring more compact configurations that facilitate planning and on-site supervision activities. Importantly, this approach reflects the impact of travel effort without imposing hard constraints on the assignment structure.

In addition, a supervision capacity constraint is introduced by defining a maximum number of students or internships that each professor can supervise. This limitation prevents excessive teaching workloads and ensures that the resulting solutions remain organizationally feasible, particularly in scenarios where the number of students significantly exceeds the number of available professors.

Finally, the contribution of academic supervisors is integrated into the objective function as an additional weighted term. The relative importance of this component can be adjusted with respect to the terms associated with students and companies, allowing the model to balance organizational criteria against the objectives of the other agents involved in the allocation process.

3 Professors Modeling

This section describes how the professors are incorporated into the SIPA formulation. The original SIPA model components (sets, parameters, decision variables, constraints, and objective terms) are assumed to remain valid unless explicitly modified. The goal is to capture supervision feasibility and organizational effort by introducing additional decision variables, parameters, constraints, and an additional component in the objective function. Although the original SIPA model is an ILP, the inclusion of aggregated supervision effort may induce non-linearities depending on the chosen distance aggregation scheme.

3.1 Additional Sets and Parameters

Let T denote the set of teachers, and let $\mathcal{E}$ denote the set of educational centers. Moreover, each professor $t \in T$ belongs to exactly one educational center, denoted by:

$$e(t) \in \mathcal{E}.$$

For each center $e \in \mathcal{E}$, define the subset of its professors:

$$T_e = \{t \in T \mid e(t) = e\}.$$

The mapping $e : T \to \mathcal{E}$ is a fixed input parameter of the model and is not part of the decision process.

For each professor $t \in T$, let:

- C_t be the maximum number of internships that professor t can supervise according to their available teaching hours.
- γ be the weight associated with the professors' contribution in the overall objective (analogous to α and β in the original SIPA objective weighting).

On the other hand, let:

- δ be the weight associated with the educational centers' contribution in the overall objective.

To model supervision-related spatial dispersion and travel effort, we assume a precomputed and normalized distance matrix:

- d_{uv}: normalized distance between two locations u and v, used to model supervision effort. Distances are assumed to be given by a precomputed distance matrix, where each location belongs to the set of relevant points composed of professors' educational centers and internship locations. In particular,

  $$u, v \in \{o_t \mid t \in T\} \cup P,$$

  where o_t denotes the educational center (origin) of professor t and each internship placement $j \in P$ is associated with a geographic location. Distances are assumed to be scaled to the interval $[0, 1]$ (e.g., through min–max normalization over the set of relevant point pairs), so that supervision-related distance terms are comparable across instances.

In the original SIPA model, a single spatial representation is assumed for all agents, so that distances between points on the map are defined in a homogeneous manner. However, the way in which these distances are used differs depending on the agent under consideration. For students, distance is employed as an individual cost associated with commuting between their place of residence and the internship location. In contrast, for professors, distance is used in an aggregated manner to evaluate the spatial dispersion of the set of internships assigned to a given supervisor.

For this reason, it is convenient to maintain both uses of the distance term—this supervision-oriented definition and the one originally employed in SIPA—despite their similar geometric interpretation.

The distance matrix underlying the values d_{uv} is constructed as a preprocessing step prior to solving the allocation problem, based on the available geographic information. Distances may be computed using Euclidean metrics, road network distances, or estimated travel times, depending on the desired level of modeling detail. The subsequent normalization to the interval $[0, 1]$ ensures that supervision-related distance measures are comparable across different instances and geographic scales, and prevents the absolute magnitude of distances from dominating the objective function. This preprocessing preserves the relative spatial structure of the problem while facilitating the integration of aggregated supervision criteria with other normalized utility terms.

3.2 Additional Decision Variables

To link students, placements, and supervisors, we update the binary decision variable:

$$x_{ijt} = \begin{cases} 1 & \text{if student } i \text{ is assigned to placement } j \text{ under the supervision of professor } t, \\ 0 & \text{otherwise,} \end{cases} \quad \forall i \in S,\ \forall j \in P,\ \forall t \in T. \tag{1}$$

3.3 Derived Quantities for Aggregated Supervision Effort

In order to evaluate supervision effort at the professor level, we define the following derived quantities, which are used to construct aggregated distance and dispersion measures.

$\mathbf{A}_t$: for each professor $t \in T$, let A_t denote the set of locations that must be covered:

$$A_t = \{o_t\} \cup \left\{ j \in P : \sum_{i \in S} x_{ijt} = 1 \right\}, \qquad \forall t \in T. \tag{2}$$

$\mathbf{L}_t$: denote the length of the minimum spanning tree over the nodes in A_t, which provides a compact proxy for spatial dispersion without explicitly defining supervision routes:

$$L_t = \sum_{(u,v)\in \mathrm{MST}(A_t)} d_{uv}, \qquad \forall t \in T. \tag{3}$$

$\mathbf{N}_t$: denote the supervision load of professor t, defined as the number of supervised students (or internships):

$$N_t = \sum_{i\in S} \sum_{j\in P} x_{ijt}, \qquad \forall t \in T. \tag{4}$$

$\mathbf{p}_t$: for each professor $t \in T$, define the normalized workload proportion as:

$$p_t = \frac{N_t}{C_t}, \qquad \forall t \in T. \tag{5}$$

$\bar{p}_e$: for each educational center $e \in \mathcal{E}$, define its global workload proportion as:

$$\bar{p}_e = \frac{1}{|T_e|} \sum_{t\in T_e} p_t, \qquad \forall e \in \mathcal{E}. \tag{6}$$

σ_e : for each educational center $e \in \mathcal{E}$, define the standard deviation of workload proportions:

$$\sigma_e = \sqrt{\frac{1}{|T_e|} \sum_{t\in T_e} (p_t - \bar{p}_e)^2}, \qquad \forall e \in \mathcal{E}. \tag{7}$$

$\bar{L}_t$**:** the average supervision dispersion cost per professor is defined as:

$$\bar{L}_t = \begin{cases} \dfrac{L_t}{N_t}, & \text{if } N_t > 0, \\ 0, & \text{if } N_t = 0, \end{cases} \qquad \forall t \in T. \tag{8}$$

Since $d_{uv} \in [0, 1]$ and the minimum spanning tree over $|A_t| = 1 + N_t$ nodes contains at most N_t edges, each bounded above by 1, it follows that $L_t \leq N_t$.

Therefore, for $N_t > 0$,

$$\bar{L}_t = \frac{L_t}{N_t} \leq 1,$$

and consequently $\bar{L}_t \in [0, 1]$. When $N_t = 0$, we define $\bar{L}_t = 0$, since no supervision effort is incurred.

Under the current definitions, the inclusion of $\bar{L}_t$ and σ_e introduces nonlinear dependencies, yielding a mixed-integer non-linear formulation.

3.4 Additional Constraints

The introduction of academic supervisors requires redefining the original SIPA constraints in order to account for the extended decision variable x_{ijt}. While the conceptual role of these constraints remains unchanged, their formulation is adapted to ensure consistency with the student–placement–professor assignment structure. In what follows, constraints (R1)–(R6) correspond to adapted versions of the original SIPA model, whereas constraint R7 arise specifically from the incorporation of professors.

(R1) Complete assignment of students. Each student must be assigned to exactly one internship placement under the supervision of exactly one professor:

$$\sum_{j \in P} \sum_{t \in T} x_{ijt} = 1, \qquad \forall i \in S. \tag{9}$$

(R2) Placement capacity. Each internship placement can be assigned to at most one student:

$$\sum_{i \in S} \sum_{t \in T} x_{ijt} \leq 1, \qquad \forall j \in P. \tag{10}$$

(R3) Maximum company quota. Each company $k \in B$ can be assigned at most Q_k students:

$$\sum_{i \in S} \sum_{j \in P_k} \sum_{t \in T} x_{ijt} \leq Q_k, \qquad \forall k \in B. \tag{11}$$

(R4) Minimum company quota. Each company $k \in B$ must be assigned at least q_k students:

$$\sum_{i \in S} \sum_{j \in P_k} \sum_{t \in T} x_{ijt} \geq q_k, \qquad \forall k \in B. \tag{12}$$

(R5) Interdependent placements (upper bound). Placements belonging to the same interdependency group must be assigned jointly:

$$\sum_{i \in S} \sum_{j \in I} \sum_{t \in T} x_{ijt} \leq |I| \, \delta_I, \qquad \forall I \in \mathcal{I}. \tag{13}$$

(R6) Interdependent placements (lower bound). If an interdependency group is activated, all its placements must be assigned:

$$\sum_{i \in S} \sum_{j \in I} \sum_{t \in T} x_{ijt} \geq |I| \, \delta_I, \qquad \forall I \in \mathcal{I}. \tag{14}$$

(R7) Maximum supervision capacity. Each professor can supervise at most C_t students or internships:

$$\sum_{i \in S} \sum_{j \in P} x_{ijt} \leq C_t, \qquad \forall t \in T. \tag{15}$$

3.5 Objective Function Extension

The incorporation of academic supervisors and educational centers introduces additional organizational criteria that must be considered in the optimization process. While the original SIPA objective function captures the preferences and objectives associated with students and companies, the extended model also accounts for the geographical compactness of supervision assignments and the proportional workload balance within each educational center.

Extended objective function. The objective function combines the four perspectives through a weighted sum, maximizing student and placement utilities while penalizing supervision dispersion and workload imbalance:

$$\max \; \alpha \sum_{i \in S} \sum_{j \in P} \sum_{t \in T} \tau_{ij} \, x_{ijt} \; + \; \beta \sum_{i \in S} \sum_{j \in P} \sum_{t \in T} \eta_{ij} \, x_{ijt} \; - \; \gamma \sum_{t \in T} \bar{L}_t \; - \; \delta \sum_{e \in \mathcal{E}} \sigma_e. \tag{16}$$

where $\tau_{ij} \in [0, 1]$ denotes the utility of assigning student i to placement j, and $\eta_{ij} \in [0, 1]$ denotes the utility of placement j (or its associated company) when assigned student i.

The weighting parameters α, β, γ and δ control the relative importance of the agents and are required to satisfy:

$$\alpha + \beta + \gamma + \delta = 1, \qquad \alpha, \beta, \gamma, \delta \in [0, 1]. \tag{17}$$